



Profile

AI engineer and medical imaging researcher with a computer science background, focusing on medical image analysis, vision-language models, LLMOps, and AI-based SaaS systems. My work connects research-driven model development with production data pipelines, workflow automation, and enterprise AI transformation.

Education

Sangmyung University

Mar. 2023 - Aug. 2026

Bachelor's Program in Computer Science / Software

- Built a strong foundation in machine learning, deep learning, data preprocessing, algorithms, and software engineering.
- Applied data-driven modeling methods to practical problems through structured preprocessing, feature engineering, and model evaluation workflows.
- Developed programming and system design skills for building reliable AI and data-intensive applications.

Research Experience

Medical Imaging Lab, Seoul National University

Sep. 2025 - Present

Research Assistant

- Advisor: Prof. Nam-Joon Kim (SNU) and Prof. Wonjae Lee, Radiologist (Samsung Medical Center).
- Participate as a member of the SUDAL Medical Imaging Team, contributing to weekly journal clubs and collaborative research discussions.
- Built the official IMSI Lab website from the ground up.
- Conduct research on conditional abdominal MRI synthesis using segmentation-guided ControlNet pipelines.
- Collaborate with Samsung Medical Center radiologists and NVIDIA researchers on medical AI projects.

Professional Experience

INTWEEN

Jun. 2026 - Present

AI SaaS / AX Engineer

- Develop in-house AI-based SaaS products, including Sasure, with GDSK and other internal service lines.
- Design pipelines to collect, clean, normalize, and integrate fragmented internal business data.
- Build workflow automation and internal AI systems that support company-wide AI transformation.
- Lead the AI transformation strategy for INTWEEN, a MICE-centered business with a revenue-driven operating model.

GDSK

Jan. 2025 - Aug. 2025

Data / AI Engineer

- Designed ETL processes that parse unstructured address text and convert it into standardized formats.
- Built stable database architecture capable of handling high-volume geographic and address-based traffic.
- Integrated fragmented public datasets and geospatial data to create feature sets for regional income estimation models.
- Developed data pipelines for collecting, preprocessing, and serving large-scale geospatial information.

OUTTA

Jul. 2024 - Present

Medical Imaging Team Lead

- Contribute to an AI organization connected to Seoul National University research and education communities.
- Design bootcamp curriculum and hands-on learning materials for Python, deep learning, and medical AI topics.
- Lead structured paper study sessions on medical image analysis, diffusion models, MAE-based representation learning, and VLMs.
- Coordinate research discussions and project execution across junior members to strengthen reproducible AI research practice.

Research Collaborations

SUDAL Medical Imaging Team

NVIDIA Collaboration

Medical Imaging AI Research

- Collaborate with Prof. Nam-Joon Kim and medical AI researchers on clinically oriented medical imaging projects.
- Work with NVIDIA researcher Ping Shu Ho (Cliff) on diffusion-based medical imaging research topics.
- Focus on medical image synthesis, reconstruction, segmentation-aware generation, and robust visual representation learning.

Key Projects

3D Latent Diffusion for MR Imaging

2026

Label-guided synthetic MRI and hepatic structure segmentation

- Contributed to research on high-resolution synthetic MR imaging using label-guided 3D latent diffusion models.
- Focused on improving data generation quality for hepatic structure segmentation workflows.

Medical Foundation Models with MAE and Mamba

2026

Self-supervised medical image learning and anatomical structure estimation

- Worked on MAE-based medical image representation learning for robust downstream performance.
- Explored Mamba-based architectures for hierarchical anatomical estimation and holistic evaluation.

Medical Vision-Language Model Development

2026

Instruction tuning for anatomy-aware VLMs

- Developed research workflows for fine-grained boundary instruction learning in anatomical vision-language models.
- Connected medical imaging tasks with multimodal reasoning, report-aligned supervision, and clinical visual understanding.

Publications

3D-LLDM: Label-Guided 3D Latent Diffusion Model for Improving High-Resolution Synthetic MR Imaging in Hepatic Structure Segmentation

K. Kim, J. Bae, Y. Han, J. Y. Bae, S. Ju, **J. Lim**, G. Kim, N. J. Kim, W. K. Jeong, et al.

2026 IEEE 23rd International Symposium on Biomedical Imaging (ISBI), 1-5 · 2026

MATHENA: Mamba-based Architectural Tooth Hierarchical Estimator and Holistic Evaluation Network for Anatomy

K. Kim, J. Park, Y. Han, A. Jung, S. Park, S. Lee, J. Yang, J. Han, S. Lee, **J. Lim**, et al.

arXiv preprint arXiv:2604.00537 · 2026

MAESIL: Masked Autoencoder for Enhanced Self-supervised Medical Image Learning

K. Kim, H. Jung, Y. Han, **J. Lim**, Y. Jean, S. Park, E. Choi, H. Go, S. Ju, S. Park, et al.

2026 International Conference on Electronics, Information, and Communication · 2026

NEMESIS: Noise-suppressed Efficient MAE with Enhanced Superpatch Integration Strategy

K. Kim, H. Jung, Y. Han, H. Go, E. Choi, S. Park, **J. Lim**, J. Yang, S. Lee, et al.

arXiv preprint arXiv:2604.01612 · 2026

MINTI: Fine Boundary Instructional Learning for Intelligent Anatomical Vision-Language Models

H. Lee, Y. Han, K. Kim, **J. Lim**, I. Um, H. Go, E. Choi, K. Kim, S. Park, J. Jeong, et al.

대한전자공학회 학술대회, 950-953 · 2026

Certifications & Licenses

TOEIC

935

Oct. 2022

Big Data Analysis Engineer

Certification

Acquired

SQLD

SQL Developer

Acquired

AWS SAA

Solutions Architect Associate

Acquired

NCA

Naver Cloud Platform

Acquired

Information Processing Engineer

National Technical Qualification

Acquired

Technical Skills

Programming Languages: Python, SQL, JavaScript, TypeScript, Java, Bash

AI / ML Frameworks: PyTorch, MONAI, Hugging Face Transformers, scikit-learn, TensorFlow

Medical AI: Medical Image Analysis, 3D Latent Diffusion, Masked Autoencoders, Mamba, Vision-Language Models, Segmentation-aware Generation

LLM / LLMOps: LLM Fine-tuning, RAG, Prompt Engineering, Model Evaluation, Workflow Automation

Data Engineering: ETL Pipelines, Data Cleaning, Geospatial Data Processing, Feature Engineering, Database Architecture

Backend & Infrastructure: FastAPI, Node.js, REST APIs, Docker, Linux, AWS, Naver Cloud Platform

Tools: Git, GitHub, VS Code, PyCharm, Docker Compose, Notion, Figma

Contact Information